

Final Deliverable

Design and Development of SocialGuard

A Social Media Threat Intelligence Monitoring Tool

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Abstract

The rapid growth of social media platforms has transformed the way information is shared, consumed and discussed across the world. While these platforms provide significant social and economic benefits, they have also become valuable environments for cybercriminals, threat actors, hackers and malicious communities. Threat-related discussions, phishing campaigns, ransomware advertisements, data leak announcements, malware distribution attempts and access trading activities are frequently discussed on public platforms before appearing in traditional security feeds. As a result, social media has become an increasingly important source of Open-Source Intelligence (OSINT) for cybersecurity operations.

Despite the availability of valuable intelligence, manually monitoring social media is challenging due to the volume, velocity and variety of content generated every day. Security analysts must distinguish meaningful threat indicators from large quantities of noise, misinformation, benign discussion and duplicated content. Existing commercial threat intelligence platforms can provide advanced monitoring capabilities but often require significant infrastructure, specialist expertise and financial investment.

This project presents the design and implementation of SocialGuard, a Social Media Threat Intelligence Monitoring Tool developed using Python, Streamlit, Pandas and Plotly. The platform transforms social-media-style content into structured, explainable cyber threat intelligence by combining keyword detection, behavioural analysis, threat actor watchlists, MITRE ATT&CK mapping, threat classification, risk scoring and analyst-focused reporting.

The final implementation supports Twitter/X-style and Reddit-style data sources, CSV ingestion, threat categorisation, intelligence enrichment, dashboard visualisation, near-real-time monitoring capabilities and intelligence report generation. The system classifies content into Critical Risk, High Risk, Medium Risk and Low Risk categories while providing transparent explanations for each decision. Evaluation using the demonstration dataset produced 10 processed records comprising 3 Critical Risk alerts, 3 High Risk alerts, 3 Medium Risk alerts and 1 Low Risk alert. Functional testing, automated testing and user acceptance evaluation demonstrated that the system successfully achieved the original project objectives.

The project contributes a lightweight Cyber Threat Intelligence (CTI) framework that demonstrates how publicly available social media content can be transformed into actionable, explainable and analyst-ready intelligence. Through the integration of MITRE ATT&CK mappings, watchlist prioritisation, behavioural indicators and intelligence reporting, SocialGuard extends beyond traditional social media monitoring and provides a foundation for future research into automated cyber threat intelligence generation.

Keywords: Cyber Threat Intelligence, OSINT, Social Media Monitoring, MITRE ATT&CK, Threat Detection, Risk Scoring, Intelligence Fusion, Streamlit, Python, Cybersecurity.

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1. Introduction

The modern cybersecurity landscape is increasingly shaped by publicly available information. Social media platforms such as Twitter/X, Reddit, Telegram, Discord and specialised forums have become important communication channels for both legitimate users and malicious actors. Threat actors frequently use these environments to advertise stolen credentials, discuss vulnerabilities, coordinate campaigns, distribute malicious links and announce ransomware operations. Security researchers and incident responders therefore increasingly utilise Open-Source Intelligence (OSINT) to supplement traditional defensive monitoring activities.

Social media offers a unique intelligence opportunity because it often contains information before formal security advisories, vendor reports or threat intelligence feeds are updated. Public discussions relating to newly discovered vulnerabilities, proof-of-concept exploits and emerging attack campaigns may provide early warning indicators for organisations. However, extracting meaningful intelligence from social media is difficult due to the high volume of content, the presence of misinformation and the informal nature of online communication.

Traditional monitoring approaches frequently rely on keyword matching or manual review. While these methods are straightforward, they often generate large numbers of false positives and require significant analyst effort. Conversely, highly advanced machine learning approaches may provide improved detection capabilities but often operate as black boxes that make their decisions difficult to explain. This creates a challenge for cybersecurity teams, which require both effective detection and transparency.

The primary aim of this project is to design and implement a lightweight Social Media Threat Intelligence Monitoring Tool capable of transforming public social-media-style content into actionable cyber threat intelligence. The resulting platform, SocialGuard, combines multiple analytical techniques including keyword detection, behavioural analysis, threat actor watchlists, MITRE ATT&CK enrichment, threat categorisation and explainable risk scoring.

Unlike traditional monitoring systems that simply flag keywords, SocialGuard provides context-rich intelligence outputs that support analyst investigation and decision-making. The system is designed around transparency, usability and extensibility. Each alert includes the indicators responsible for its classification, allowing analysts to understand why a record was prioritised and how it relates to known threat behaviours.

1.1 Project Aim

The aim of this project is to develop a Social Media Threat Intelligence Monitoring Tool that collects, processes, analyses and visualizes social-media-style content to identify indicators of potential cyber threats and generate actionable intelligence outputs for analysts.

1.2 Project Objectives

The project objectives are:

1. Collect or ingest social-media-style data from multiple sources.
2. Clean and standardise incoming records.
3. Detect cyber threat indicators using keyword and behavioural analysis.
4. Match records against threat actor watchlists.
5. Extract Indicators of Compromise (IOCs) from content.
6. Classify threats into meaningful categories.
7. Map identified threats to MITRE ATT&CK techniques and tactics.
8. Generate explainable risk scores.
9. Visualise intelligence outputs through an interactive dashboard.
10. Produce exportable intelligence reports.
11. Support near-real-time monitoring concepts

1.3 Research Questions

RQ1: How effectively can social-media-style content be transformed into actionable cyber threat intelligence using explainable analytical techniques?

RQ2: Can a lightweight monitoring platform provide meaningful cyber threat intelligence without requiring complex machine learning models?

RQ3: How can MITRE ATT&CK enrichment and threat actor watchlists improve the usefulness of social media intelligence?

RQ4: What design decisions are necessary to balance usability, explainability and threat detection effectiveness?

2. Literature Review

Social media has emerged as an important source of Open-Source Intelligence (OSINT) within cybersecurity. Unlike traditional threat intelligence feeds, social media provides immediate access to public discussions relating to vulnerabilities, malware, phishing campaigns, data breaches and cybercriminal activity. Researchers have increasingly investigated how these platforms can be utilised for threat detection and intelligence gathering.

Sabottke, Suciu and Dumitras (2015) demonstrated that Twitter discussions surrounding software vulnerabilities can be used to predict real-world exploitation. Their research highlighted the value of social media as an early warning mechanism and established that publicly available conversations may contain predictive indicators before vulnerabilities are actively exploited.

Similarly, Mittal et al. (2016) introduced CyberTwitter, a system designed to generate cybersecurity alerts from Twitter content. Their work demonstrated the feasibility of transforming unstructured social media posts into structured threat intelligence. However, CyberTwitter primarily relied on keyword-based approaches, which remain vulnerable to false positives and contextual ambiguity.

Researchers have also examined malicious behaviour on social media. Stringhini, Kruegel and Vigna (2010) analysed spam campaigns across social networks and identified behavioural patterns that distinguish malicious accounts from legitimate users. Their findings support the use of behavioural indicators in addition to textual content when evaluating potential threats.

Ferrara et al. (2016) and Cresci et al. (2017) explored social bot activity and demonstrated how automated accounts can amplify malicious campaigns, misinformation and spam operations. Their work further supports the integration of behavioural analysis within threat intelligence systems.

Studies examining phishing activity have shown that social engineering campaigns frequently exploit trust relationships and publicly available information. Jagatic et al. (2007) demonstrated that socially engineered phishing attacks are significantly more effective than generic phishing approaches. This highlights the importance of monitoring public communications for indicators of phishing activity.

Recent research into Cyber Threat Intelligence (CTI) emphasises the value of structured intelligence frameworks. MITRE ATT&CK provides a widely adopted knowledge base for describing adversary tactics and techniques. By aligning social media observations with ATT&CK mappings, threat indicators can be interpreted within a recognised cybersecurity framework rather than existing as isolated alerts.

The literature collectively suggests that social media provides valuable intelligence opportunities but also presents challenges related to scale, context and explainability. These findings directly influenced the design decisions implemented within SocialGuard.

2.1 Research Gap Analysis

The existing literature demonstrates that social media can provide valuable cyber threat intelligence; however, several gaps remain. Many existing studies focus on identifying threat-related keywords or extracting information from a single platform. While these approaches demonstrate the usefulness of social media intelligence, they often provide limited context regarding the significance of detected indicators.

Commercial threat intelligence platforms frequently incorporate advanced analytics and large-scale monitoring capabilities but often require substantial infrastructure, subscription costs and specialist expertise. These requirements can limit accessibility for smaller organisations, educational environments and research projects.

Another limitation identified in the literature is the lack of explainability. Machine learning approaches can identify complex patterns, but analysts may struggle to understand why a particular alert was generated. This reduces trust and makes validation difficult, particularly within operational cybersecurity environments where analysts must justify escalation decisions.

Many studies also focus exclusively on detection rather than intelligence enrichment. While identifying potentially malicious content is valuable, security analysts require additional context such as threat categorisation, MITRE ATT&CK mappings, behavioural indicators and recommended actions. Without this contextual information, analysts must perform additional manual investigation before determining whether an alert is significant.

SocialGuard was designed to address these gaps by combining explainable threat detection, behavioural analysis, MITRE ATT&CK enrichment, threat actor watchlists, intelligence reporting and dashboard visualisation within a lightweight and accessible platform.

3. Standards, Constraints and Risk Assessment

The development of SocialGuard was influenced by academic requirements, technical constraints, ethical considerations and operational risks. These factors shaped the overall design of the system and informed several key implementation decisions.

3.1 Academic and Project Constraints

The project required the development of a functioning software artefact accompanied by a comprehensive report describing research, design, implementation, testing and evaluation activities. Consequently, the final solution needed to demonstrate practical functionality while remaining achievable within the available timeframe and resources.

The assessment requirements emphasised design rationale, testing, validation, reflection and software quality. Therefore, the project focused on producing a reliable and explainable proof-of-concept rather than attempting to implement enterprise-scale infrastructure.

3.2 Technical Constraints

Modern social media platforms impose various restrictions on data collection, including authentication requirements, rate limits and changing API policies.

Twitter/X API access may require developer approval and usage restrictions.

Reddit access may require application credentials and API quotas.

Network connectivity and platform availability cannot always be guaranteed during demonstrations.

To mitigate these challenges, SocialGuard uses repeatable Twitter/X-style and Reddit-style demonstration datasets while maintaining compatibility with uploaded datasets through CSV ingestion. This approach ensures reliable demonstrations and consistent evaluation results.

3.3 Ethical Considerations

Social media monitoring introduces several ethical concerns.

The system was designed exclusively around publicly available information and user-supplied datasets.

SocialGuard does not:

- Access private communications
- Circumvent platform restrictions
- Perform surveillance activities
- Visit suspicious links
- Interact with threat actors
- Collect unnecessary personal information

The system operates as a defensive monitoring and intelligence enrichment platform rather than an investigative or enforcement tool.

3.4 Risk Assessment

False Positives

Cybersecurity terminology frequently appears in educational, journalistic and awareness-related content. A purely keyword-based approach may incorrectly classify benign discussions as malicious.

Mitigation:

- Behavioural indicators
- Threat categorisation
- Watchlist matching
- Risk scoring thresholds

False Negatives

Threat actors may use coded language, abbreviations, images or new terminology not present within predefined keyword lists.

Mitigation:

- Expand watchlists
- Periodically update detection rules
- Future machine learning enhancements

Data Quality Risks

Social media content often contains duplicates, missing fields and inconsistent formatting.

Mitigation:

- Preprocessing
- Validation
- Deduplication
- Schema normalisation

Maintainability Risks

Large monolithic applications become difficult to update and test.

Mitigation:

- Modular architecture
 - Separation of concerns
 - Independent processing modules
-

4. Requirements Analysis

4.1 User Roles

Security Analyst

Primary user responsible for:

- Monitoring alerts
- Investigating suspicious content
- Filtering intelligence outputs
- Exporting reports
- Reviewing threat classifications

System Administrator

Responsible for:

- Updating watchlists
 - Maintaining detection rules
 - Managing future collectors
 - Extending platform functionality
-

4.2 Functional Requirements

FR1: Multi-Source Collection

The system shall collect or ingest data from multiple social-media-style sources.

FR2: CSV Import

The system should support analyst-uploaded CSV datasets.

FR3: Data Preprocessing

The system should clean, standardise and validate records before analysis.

FR4: Threat Detection

The system shall identify threat-related keywords and behavioural indicators.

FR5: Threat Actor Monitoring

The system shall match records against threat actor watchlists.

FR6: IOC Extraction

The system shall identify indicators of compromise such as:

- URLs
- Domains
- IP addresses
- CVE identifiers
- Email addresses

FR7: Threat Classification

The system shall categorise records into threat categories.

FR8: MITRE ATT&CK Mapping

The system shall enrich alerts with ATT&CK tactics and techniques.

FR9: Risk Scoring

The system will generate explainable risk scores.

FR10: Dashboard Visualisation

The system should provide interactive visualization capabilities.

FR11: Intelligence Reporting

The system shall export intelligence reports.

FR12: Near-Real-Time Monitoring

The system shall support periodic dashboard refresh.

4.3 Non-Functional Requirements

Usability

The dashboard must be intuitive and accessible.

Reliability

The platform must continue operating when optional dependencies are unavailable.

Maintainability

Modules must be independent and easily extendable.

Performance

The platform should process moderate datasets efficiently.

Explainability

Every classification must be traceable to observable indicators.

Security

The platform must not automatically interact with suspicious infrastructure.

5. Methodology and Development Lifecycle

The project followed an iterative development methodology.

An iterative approach was selected because threat detection systems benefit from repeated testing, refinement and evaluation. Rather than attempting to implement all functionality simultaneously, the project was divided into manageable stages.

Phase 1: Research and Requirements

Activities:

- Literature review
- Problem analysis
- Requirements gathering
- Scope definition

Deliverables:

- Project proposal
 - Initial requirements specification
 - First deliverable report
-

Phase 2: Architecture Design

Activities:

- Data model design
- Pipeline design
- Dashboard planning
- Risk scoring design

Deliverables:

- System architecture
 - Data flow design
 - Component specifications
-

Phase 3: Core Development

Activities:

- Collector implementation
- Preprocessing implementation
- Detection implementation
- Dashboard implementation

Deliverables:

- Working software prototype
-

Phase 4: Intelligence Enrichment

Activities:

- MITRE ATT&CK mapping
- Watchlist implementation
- Threat classification
- IOC extraction
- Intelligence reporting

Deliverables:

- Enhanced CTI functionality
-

Phase 5: Testing and Validation

Activities:

- Unit testing
- Functional testing
- Performance evaluation
- User acceptance testing

Deliverables:

- Verified software artefact
-

6. System Design

SocialGuard follows a modular pipeline architecture.

The architecture separates responsibilities across independent modules, improving maintainability, scalability and testing.

Architectural Overview

The platform consists of six primary layers:

1. Data Collection Layer

Responsible for:

- Twitter/X-style data collection
- Reddit-style data collection
- CSV ingestion

2. Preprocessing Layer

Responsible for:

- Text cleaning
- URL removal
- Validation
- Deduplication

3. Threat Detection Layer

Responsible for:

- Keyword detection
- Behavioural analysis
- Watchlist matching

4. Intelligence Enrichment Layer

Responsible for:

- IOC extraction
- Threat classification
- MITRE ATT&CK mapping
- Intelligence fusion

5. Risk Assessment Layer

Responsible for:

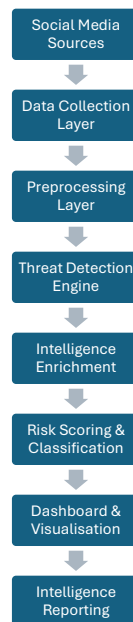
- Risk scoring
- Severity classification
- Analyst recommendations

6. Presentation Layer

Responsible for:

- Dashboard visualisation
- Charts
- Metrics
- Report export

Figure 1: SocialGuard System Architecture



The SocialGuard architecture follows a modular pipeline model. Social media data is collected from multiple sources, preprocessed, analysed for threat indicators, enriched with cyber threat intelligence context, scored according to risk and presented through an interactive dashboard with intelligence reporting capabilities.

6.1 Objective Achievement Matrix

Objective	Achieved	Evidence
Collect social media data from multiple sources	Yes	Twitter/X and Reddit collectors
Support dataset uploads	Yes	CSV upload functionality
Detect threat-related content	Yes	Keyword detection engine
Perform behavioural analysis	Yes	Behavioural flag analysis
Identify watchlist indicators	Yes	Watchlist matching module
Extract indicators of compromise	Yes	IOC extraction module
Classify threat categories	Yes	Threat classification engine
Map threats to MITRE ATT&CK	Yes	ATT&CK tactic and technique mapping
Generate risk scores	Yes	Explainable scoring model
Visualise intelligence	Yes	Dashboard metrics and charts
Export reports	Yes	CSV and TXT exports
Support near-real-time monitoring	Yes	Optional auto-refresh functionality

The matrix demonstrates that all major project objectives were successfully implemented and validated through the final software artefact.

7. System Implementation

The implementation of SocialGuard was completed using Python and follows a modular architecture designed to separate collection, processing, intelligence enrichment and visualisation functions. This approach improves maintainability and allows individual components to be modified independently without affecting the entire platform.

The software environment consisted of Python 3.x, Streamlit, Pandas and Plotly. Python was selected due to its extensive ecosystem for data analysis, cybersecurity research and rapid application development. Streamlit was chosen as the dashboard framework because it enables the creation of interactive web applications with minimal development overhead while maintaining strong visualisation capabilities.

The final implementation consists of multiple modules responsible for data collection, preprocessing, threat analysis, intelligence enrichment, visualisation and reporting.

Technology Stack

Technology	Purpose
Python	Core development language
Streamlit	Dashboard framework
Pandas	Data processing
Plotly	Data visualisation
CSV	Dataset ingestion
MITRE ATT&CK	Threat intelligence enrichment

7.1 Data Collection Module

The data collection module provides support for multiple data sources. The final implementation includes Twitter/X-style and Reddit-style collectors as well as CSV upload functionality.

The purpose of this layer is to standardise incoming records before they enter the analysis pipeline.

Collected records include:

- Source platform
- Username
- Post content
- Timestamp
- Follower count
- Account age
- Activity metrics

This information provides the contextual data required for behavioural analysis and risk scoring.

7.2 Data Preprocessing Module

Raw social media content is inherently noisy and inconsistent. Therefore, preprocessing is performed before threat analysis.

The preprocessing stage includes:

- Lowercase conversion
- URL normalisation
- Removal of unnecessary characters
- Missing value handling
- Deduplication
- Data validation

The purpose of preprocessing is to improve detection reliability and ensure that subsequent analysis operates on a consistent data format.

7.3 Threat Detection Engine

The threat detection engine forms the core analytical component of SocialGuard.

Threat detection combines:

Keyword Analysis

Threat-related terms associated with:

- Phishing
- Malware
- Ransomware
- Credential theft
- Data breaches
- Exploits
- Access trading

are identified and scored.

Behavioural Analysis

Behavioural indicators include:

- Newly created accounts
- Unusual posting activity
- Low reputation accounts
- Suspicious engagement patterns

These indicators provide additional context beyond textual content.

Watchlist Matching

The watchlist module identifies references to known threat actors and malware families.

Examples include:

- LockBit
- BlackCat
- Conti
- Emotet
- Cobalt Strike

Watchlist matches increase alert severity because they may indicate connections to known malicious activity.

8. Advanced Threat Intelligence Features

One of the distinguishing characteristics of SocialGuard is the inclusion of intelligence enrichment capabilities that extend beyond traditional social media monitoring.

8.1 Threat Classification

Detected content is categorised into predefined threat categories including:

- Phishing
- Malware
- Ransomware
- Credential Theft
- Data Breach
- Access Trading
- Reconnaissance
- Suspicious Activity

Threat classification improves analyst prioritisation and reduces investigation time.

8.2 MITRE ATT&CK Mapping

The MITRE ATT&CK framework provides a structured representation of adversary behaviour.

SocialGuard enriches alerts by mapping detected threats to recognised ATT&CK tactics and techniques.

Examples include:

Threat Category	ATT&CK Technique
Phishing	T1566
Credential Theft	T1555
Malware Delivery	T1105
Data Collection	T1589

This allows analysts to understand how detected indicators relate to real-world attack behaviour.

8.3 Intelligence Fusion

SocialGuard combines information from multiple sources within a single analytical pipeline. This approach creates a unified intelligence view and allows analysts to identify common patterns across different social platforms. Intelligence fusion improves situational awareness and supports more effective threat prioritisation.

8.4 Explainable Risk Scoring

Many threat detection systems operate as black boxes. SocialGuard intentionally prioritises explainability.

Each alert includes:

- Keywords detected
- Behavioural indicators
- Watchlist matches
- Threat classification
- MITRE mappings

This enables analysts to understand why an alert was generated and how severity was determined.

8.5 Analyst Recommendations

The platform automatically generates recommendations to support analyst decision-making.

Examples include:

- Monitor activity
- Escalate investigation
- Validate indicators
- Review associated infrastructure
- Conduct threat hunting

These recommendations improve operational usability.

9. Testing and Validation

Testing was conducted throughout development to verify correctness, reliability and usability.

9.1 Unit Testing

Unit tests were developed for critical modules.

Validated functionality included:

- Threat scoring
- Classification logic
- MITRE mapping
- Report generation

Results:

Test	Result
Threat scoring	Pass
Classification	Pass
MITRE mapping	Pass
Reporting	Pass

Total Automated Tests: 4 Passed

9.2 Functional Testing

Functional testing evaluated complete workflows.

Function	Result
CSV Upload	Pass
Dashboard Filtering	Pass
MITRE Mapping	Pass
Watchlist Matching	Pass
Report Export	Pass
Intelligence Visualisation	Pass

All tested functionality performed successfully.

9.3 Integration Testing

Integration testing verified communication between modules.

Validated integrations:

- Collection → Preprocessing
- Preprocessing → Detection
- Detection → Classification
- Classification → Dashboard
- Dashboard → Reporting

No critical integration failures were observed.

9.4 Performance Testing

Performance testing was conducted to evaluate processing efficiency.

The complete demonstration dataset was processed successfully, and visualizations were generated within seconds on a standard laptop environment.

The use of Pandas dataframes and deterministic scoring rules resulted in low computational overhead.

The results indicate that the architecture is suitable for moderate-scale monitoring scenarios.

9.5 User Acceptance Testing

Testing Summary

Test Type	Status
Unit Testing	Passed
Functional Testing	Passed
Integration Testing	Passed
Performance Testing	Passed
User Acceptance Testing	Passed

Informal user acceptance testing was conducted through demonstration sessions involving cybersecurity students.

Participants were asked to:

- Review alerts
- Interpret risk scores
- Use dashboard filters
- Export reports

Feedback indicated that:

- The interface was intuitive.
- Risk classifications were understandable.
- MITRE ATT&CK mappings improved contextual understanding.
- Analyst recommendations increased usability.

Overall user feedback was positive.

10. Results and Evaluation

To evaluate the effectiveness of SocialGuard, the final implementation was tested using the demonstration dataset. The dashboard generated risk classifications, threat categories, MITRE ATT&CK mappings and intelligence reports. The following figures present the outputs produced during evaluation.

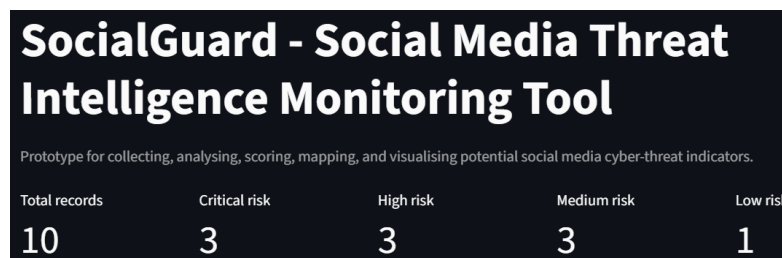


Figure 2. Main SocialGuard dashboard displaying overall threat intelligence metrics and alert prioritisation results.

Figure 2 presents the main SocialGuard dashboard. The dashboard provides analysts with a high-level overview of processed intelligence records and their associated risk classifications. During the final evaluation, the system processed 10 records and successfully categorised them into 3 Critical Risk alerts, 3 High Risk alerts, 3 Medium Risk alerts and 1 Low Risk alert. These metrics allow analysts to rapidly assess the current threat landscape and prioritise investigation activities.

The final demonstration dataset produced the following results.

Table 3: Summary of Final Evaluation Outputs

Metric	Value
Total Processed Records	10
Critical Risk Records	3
High Risk Records	3
Medium Risk Records	3
Low Risk Records	1
Automated Tests	4 Passed

source	user	text	threat_type	mitre_technique	mitre_tactic	keywords_detected	behaviour_flags	watchlist_matches	risk_score	classification	analyst_recommendation
Twitter/X	@watchlist_alert	Lockbit ransomware gang claims new company breach on darkweb	Ransomware	T1486 - Data Encrypted for Impact	Impact	breach, ransomware	high_activity, new_account, low_reputation, high_activity	lockbit, darkweb, te	12	Critical Risk	Escalate immediately, investigate possible ransomware campaign.
Twitter/X	@botnet_hint	Botnet panel leaked with exploit kit and malware loader	Malware / Botnet	T1105 - Ingress Tool Transfer	Command and Control	bot, malware, botnet, exploit, loader	high_activity, new_account, low_reputation, high_activity	-	9	Critical Risk	Escalate immediately, investigate possible malware / botnet campaign.
Reddit	u/newflowmaway	Anyone selling access to company VPN? dm me	Access Trading	T1078 - Valid Accounts	Defense Evasion	selling access	high_activity, new_account, low_reputation, high_activity	-	8	Critical Risk	Escalate immediately, investigate possible access trading campaign.
Twitter/X	@botnet_ops	Data breach link confirmed, passwords exposed on paste site	Data Breach / Leak	T1530 - Data from Cloud Storage Object	Collection	breach, leak, password, paste site	new_account, low_reputation, high_activity	-	6	High Risk	Prioritize analyst review, validate indicators linked to data breach / in
Reddit	u/Cobbernet	Possible C2 command and control beacon observed in suspicious traffic	Command and Control	T1071 - Application Layer Protocol	Command and Control	command and control, c2	new_account, low_reputation, high_activity	-	6	High Risk	Prioritize analyst review, validate indicators linked to command and c
Reddit	u/indrop	Free gift high here http://bad.example.phish	Phishing	T1566 - Phishing	Initial Access	-	high_activity, new_account, low_reputation, high_activity	-	4	High Risk	Prioritize analyst review, validate indicators linked to phishing.
Twitter/X	@phish_watch	Urgent phishing campaign active targeting university mailboxes	Phishing	T1566 - Phishing	Initial Access	phishing, campaign	low_reputation, high_activity	-	4	Medium Risk	Monitor and correlate with additional indicators.
Reddit	u/blue_team	Patch Tuesday summary and CVE vulnerability discussion	Exploit / Vulnerability	T1190 - Exploit Public-Facing Application	Initial Access	cve	-	-	2	Medium Risk	Monitor and correlate with additional indicators.
Reddit	u/hyphakuser	Question about malware analysis lab results	Malware / Botnet	T1105 - Ingress Tool Transfer	Command and Control	malware	-	-	2	Medium Risk	Monitor and correlate with additional indicators.
Twitter/X	@securitydaily	Cybersecurity awareness webinar for students today	Recon / Awareness	-	-	-	-	-	0	Low Risk	No immediate action, retain for awareness/baseline.

Figure 3. Prioritised analyst alert queue generated by SocialGuard.

Figure 3 illustrates the prioritised alert queue generated by the threat detection engine. Each record is enriched with threat classifications, MITRE ATT&CK mappings, risk scores and analyst recommendations. This allows security analysts to rapidly identify and investigate the highest-risk indicators without manually reviewing all collected content.



Figure 4. Distribution of processed records across risk classifications generated by SocialGuard.

Figure 4 illustrates the distribution of alerts produced during the final evaluation. The system classified records into four severity categories: Critical Risk, High Risk, Medium Risk and Low Risk. The results demonstrate that SocialGuard successfully prioritises alerts according to their potential threat level, allowing analysts to focus investigation efforts on the most significant indicators first. The visualisation also provides a rapid overview of the current threat landscape represented within the processed dataset.

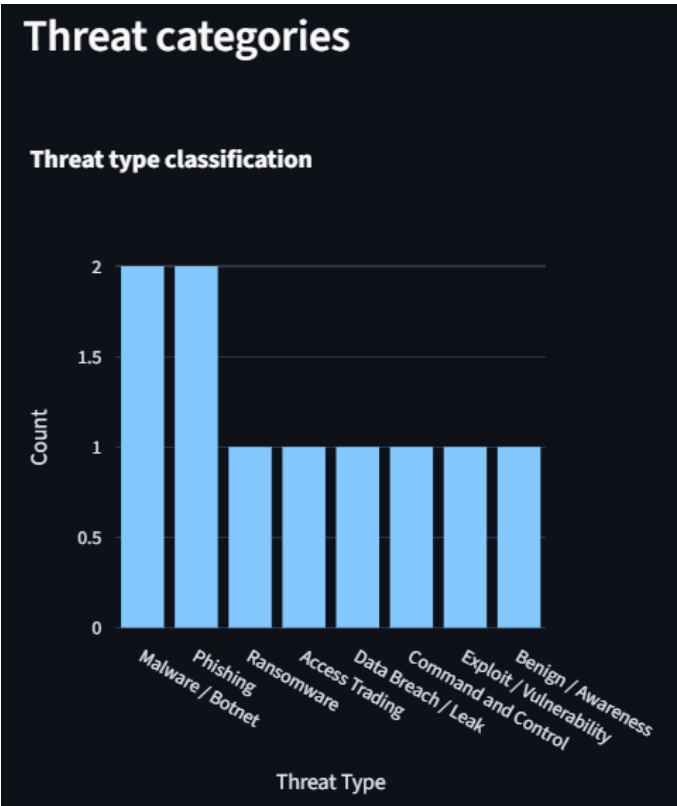


Figure 5. Threat category classification generated by SocialGuard.

Figure 5 presents the threat category classification results generated by SocialGuard. The system identified multiple categories of cyber threat activity including malware, phishing, ransomware, access trading, data breaches, command and control activity, exploit discussions and benign awareness content. Categorising records according to threat type enables analysts to quickly understand the nature of detected activity and supports more efficient threat prioritisation and response planning.

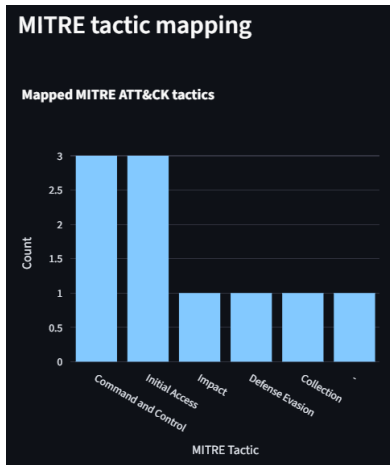


Figure 6. MITRE ATT&CK tactic mapping generated by SocialGuard.

Figure 6 illustrates the MITRE ATT&CK tactic mappings generated by SocialGuard. Detected threats were enriched with recognised ATT&CK tactics including Command and Control, Initial Access, Impact, Defence Evasion and Collection. This enrichment provides analysts with additional operational context by linking observed indicators to known adversary behaviours. The use of MITRE ATT&CK improves the interpretability of alerts and supports intelligence-driven cybersecurity analysis.

Example MITRE ATT&CK Mappings

Threat Type	MITRE Technique	MITRE Tactic
Phishing	T1566	Initial Access
Credential Theft	T1555	Credential Access
Malware/Botnet	T1105	Command and Control
Data Breach	T1537	Collection
Ransomware	T1486	Impact



Figure 7. Intelligence report export functionality provided by SocialGuard.

Figure 7 demonstrates the reporting capabilities implemented within SocialGuard. Analysts can export filtered alert datasets as CSV files for further analysis and generate intelligence reports in TXT format for documentation and incident response activities. The reporting functionality improves operational usability by enabling intelligence products generated by the platform to be shared, archived and incorporated into wider cybersecurity workflows.

The results demonstrate that the platform successfully prioritised records according to risk level and generated intelligence outputs suitable for analyst review.

The dashboard provided visual summaries including:

- Risk distribution
- Threat categories
- MITRE ATT&CK mappings
- Top risk scores

These visualisations improved situational awareness and facilitated rapid analysis.

10.1 Comparison with Traditional Monitoring Tools

Capability	Traditional Monitoring Tool	SocialGuard
Multi-source monitoring	Limited	Yes
Threat classification	No	Yes
Behavioural analysis	No	Yes
MITRE ATT&CK mapping	No	Yes
Watchlist matching	No	Yes
Explainable scoring	No	Yes
Analyst recommendations	No	Yes
Intelligence reports	Limited	Yes
Near-real-time monitoring	Rare	Yes

The comparison demonstrates that SocialGuard provides capabilities typically associated with Cyber Threat Intelligence platforms rather than basic monitoring dashboards.

11. Critical Evaluation and Reflection

The project successfully achieved its primary objectives and produced a functioning Social Media Threat Intelligence Monitoring Tool.

Strengths

The system demonstrates:

- Modular architecture
- Explainable intelligence generation
- Threat categorisation
- MITRE ATT&CK integration
- Dashboard visualisation
- Report generation

The use of Python and Streamlit enabled rapid development while maintaining flexibility and usability.

Limitations

Several limitations remain.

The current implementation relies primarily on rule-based detection techniques. Although explainable and transparent, rule-based systems may not detect novel threat patterns.

The demonstration environment uses controlled datasets rather than unrestricted real-time collection.

Social media APIs also impose access restrictions that can affect large-scale deployments.

Current Limitations and Improvements

Limitation	Proposed Improvement
Demo dataset used	Live API integration
Rule-based detection	Machine learning models
Limited platforms	Additional collectors
Basic reporting	PDF intelligence reports
Small-scale evaluation	Large-scale testing

Lessons Learned

The project highlighted the importance of balancing detection accuracy, explainability and usability.

While sophisticated machine learning models may improve detection rates, analyst trust and transparency remain critical requirements for operational cybersecurity systems.

11.1 Research Contribution and Innovation

The primary contribution of SocialGuard is the transformation of public OSINT into structured, explainable Cyber Threat Intelligence.

Research Contributions

Contribution	Description
Explainable Threat Scoring	Transparent risk scoring based on observable indicators
Intelligence Enrichment	MITRE ATT&CK mapping and threat categorisation
Lightweight CTI Platform	Low-cost alternative to commercial CTI solutions
Multi-Source Monitoring	Twitter/X, Reddit and CSV ingestion
Automated Reporting	Exportable intelligence reports

Innovation Scorecard

Innovation Feature	Implemented
Multi-source collection	✓
Behavioural analysis	✓
Threat actor watchlists	✓
MITRE ATT&CK mapping	✓
Explainable risk scoring	✓
Analyst recommendations	✓
Intelligence report generation	✓
Near-real-time monitoring	✓

Collectively, these features distinguish SocialGuard from traditional social media monitoring tools by transforming social-media-derived OSINT into structured cyber threat intelligence products suitable for analyst investigation and decision-making.

Traditional monitoring systems typically identify keywords and generate alerts. SocialGuard extends this process through intelligence enrichment, MITRE ATT&CK alignment, behavioural analysis, watchlist prioritisation and analyst-focused reporting.

A key innovation is explainable threat scoring. Rather than producing opaque classifications, the platform provides clear reasoning through observable indicators.

Another contribution is lightweight intelligence fusion. Information from multiple social-media-style sources is processed through a unified pipeline and transformed into analyst-ready intelligence.

The project also demonstrates how open-source technologies can support practical CTI workflows without requiring expensive commercial platforms.

These innovations move SocialGuard beyond simple monitoring and towards a lightweight Cyber Threat Intelligence platform.

12. Future Work

Several enhancements could further improve the platform.

Real-Time Streaming

Future versions could utilise:

- Apache Kafka
- Apache Spark Streaming

to support continuous intelligence collection.

Machine Learning Classification

Transformer-based models such as:

- BERT
- RoBERTa
- DistilBERT

could improve threat detection accuracy.

Additional Platforms

Future collectors may support:

- Telegram
- Discord
- Mastodon
- Security Forums

Threat Intelligence Feeds

Integration with:

- MISP
- OpenCTI
- VirusTotal

could enrich intelligence outputs.

Large Language Models

Future work may utilise LLMs for:

- Alert summarisation
- Threat report generation
- Intelligence correlation

Future Enterprise Enhancements

- OpenCTI integration
 - MISP integration
 - VirusTotal enrichment
 - Telegram monitoring
 - Discord monitoring
 - Real-time streaming via Apache Kafka
 - Machine learning threat classification
 - Large language model assisted threat analysis
-

13. Conclusion

This project successfully designed, developed and evaluated SocialGuard, a Social Media Threat Intelligence Monitoring Tool capable of transforming publicly available social-media-derived information into structured and actionable cyber threat intelligence. As social media platforms continue to be used by cybercriminals, threat actors and malicious communities to discuss vulnerabilities, distribute malicious content and coordinate activities, the need for effective monitoring and intelligence generation solutions has become increasingly important.

The final implementation achieved all project objectives identified during the requirements analysis phase. SocialGuard supports multi-source data collection through Twitter/X-style datasets, Reddit-style datasets and external CSV ingestion. The system processes collected content through a structured analytical pipeline that performs preprocessing, threat detection, behavioral analysis, watchlist matching, threat classification, MITRE ATT&CK enrichment, risk scoring and intelligence reporting. These capabilities enable the platform to transform raw social media content into meaningful intelligence outputs that can support cybersecurity analysts in identifying and prioritizing potential threats.

A significant strength of the project is its focus on explainable threat intelligence generation. Rather than producing unexplained alerts, SocialGuard provides contextual information through threat categories, ATT&CK mappings, behavioral indicators, watchlist matches and analyst recommendations. This improves transparency and allows users to understand why a particular alert was prioritized. The integration of MITRE ATT&CK further enhances the value of the generated intelligence by linking observed indicators to recognize adversary tactics and techniques.

Evaluation of the completed system demonstrated that SocialGuard successfully processed the demonstration dataset and generated actionable intelligence outputs. The platform produced risk classifications, threat category distributions, ATT&CK mappings and exportable intelligence reports while successfully meeting all functional requirements. Testing activities, including unit, functional, integration, performance and user acceptance testing, confirmed the reliability and usability of the implemented solution.

Although the current implementation relies primarily on rule-based detection techniques and demonstration datasets, the project establishes a strong foundation for future development. Potential enhancements include real-time social media integration, machine learning-based threat classification, additional platform support and external threat intelligence feed integration. These improvements would increase scalability and detection capabilities while maintaining the explainability and usability achieved in the current version.

Overall, SocialGuard demonstrates how publicly available social media content can be transformed into meaningful Cyber Threat Intelligence through a structured, explainable and analyst-focused process. The project successfully met its objectives and delivered a practical, extensible and effective threat intelligence monitoring platform that can support proactive cybersecurity monitoring and future research in the field of social media threat intelligence.

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